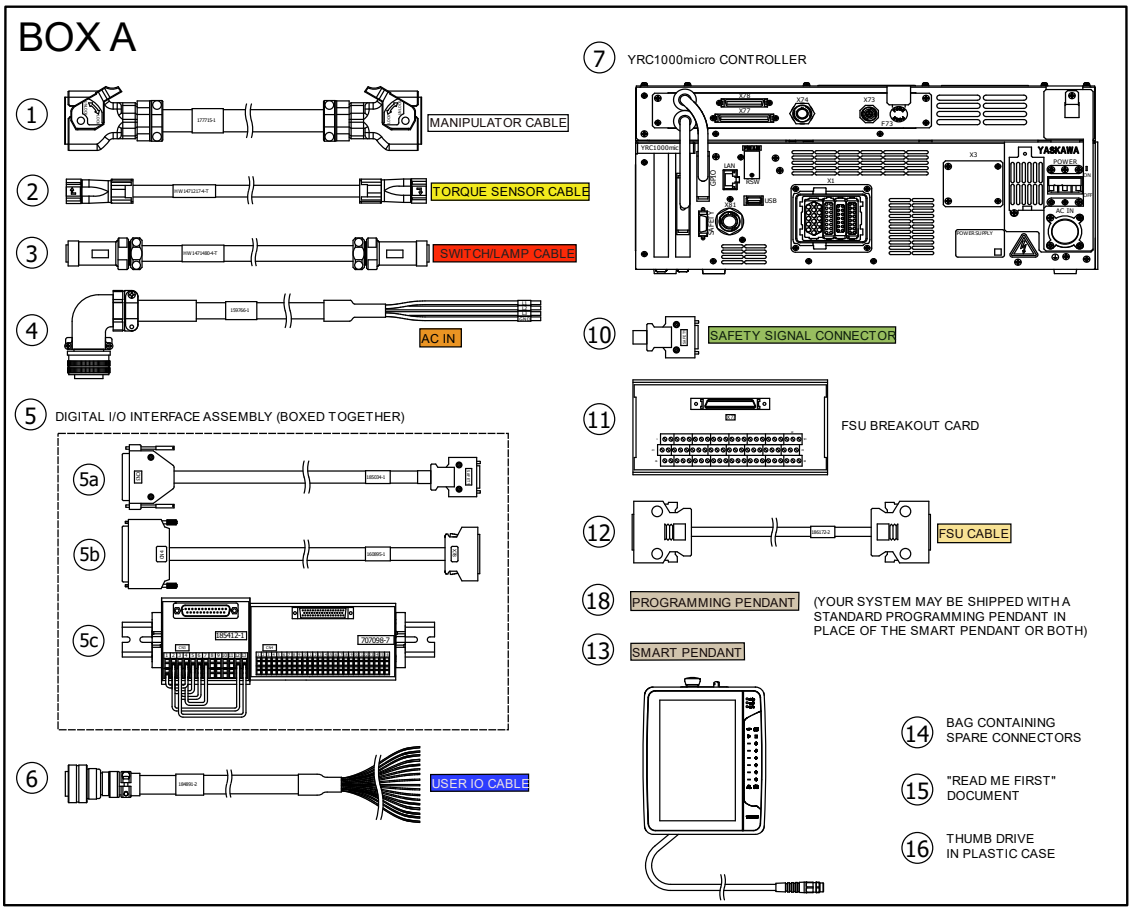
SMART SERIES ROBOT
Included Equipment

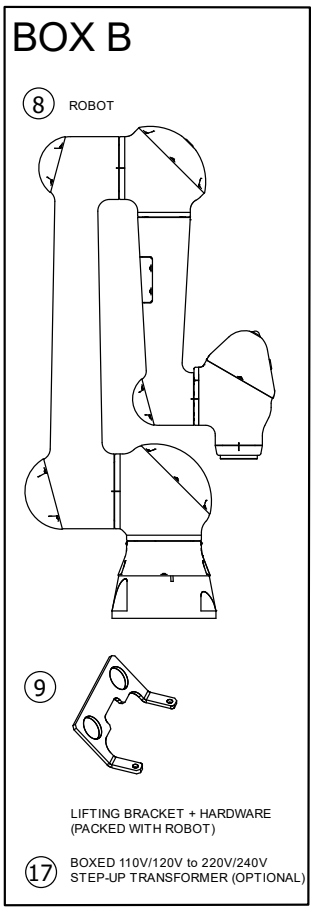
CAUTION! Use proper lifting techniques when unpacking. Not using proper techniques and/or equipment can result in personnel or equipment damage

Locate and verify the following items. If anything is missing notify your local YASKAWA representative as soon as possible.

For more information scan the QR code or go to

https://www.motoman.com/setup/hc10





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Unboxing Methods

The largest item in the shipment is the HC10DT robot. It weighs approximately 105lbs. Team lift the robot using OSHA safety guidelines, or use a crane or other lifting equipment to lift and position the robot. Lift instructions using a crane are in Section 2.1.1 (pages 2-1 and 2-2) of the HC10DT Instructions manual (HW1485083).

Mounting

Once the robot has been removed from the crate and placed into position, it is important to properly mount it to a secure base. Make sure to use appropriate hardware. The baseplate should be rugged and durable to prevent shifting of the manipulator or the mounting fixture. A baseplate at least 32 mm or thick and anchor bolts M12 or larger are recommended. More detailed information on installation options may be found in the HC10DT Instructions Manual (HW1485083).

Connecting the Robot Cables

Once your robot is securely anchored to an appropriate base, the next step is to connect the robot cables. Most of the cables are color coded to assist in making proper connections.

CAUTION: Do not remove or connect the TQ and SW connectors while the power supply of the controller is turned ON. Failure to observe this instruction may result in damage to the sensor.

Start connecting cables on the robot side. This way you can unroll the cables and route them easier than if the cables are tangled or twisted together. There are three connectors on the base of the robot: Manipulator (1BC), Switch/Lamp (SW) and Torque Sensor (TQ).

Remove shipping covers from the connectors. Insert the 1BC manipulator cable (1) into the 1BC connector (color coded white with black stripes), then tighten the lever by hand until snug. If the connecter lever will not close, make sure the connector is fully seated and try again. Once connected and locked, lay the cable across the floor toward the controller. If you need to remove the 1BC manipulator cable, there is a small safety catch at the end of the lever. Depress the catch to release the lever.

The next two cables, (2) and (3), have male and female ends. The female ends of the cables (no pins) attach to the robot. To connect the cables to the robot base, gently insert until the connector fully seats and then turn the collar on the connector the direction indicated by the arrow to secure.

The connectors are keyed so that they only fit when properly aligned. If you cannot insert the connector, rotate it slowly until the guide pin and slot align and the connector seats.

The cable for the switch/lamp (3) connects to the SW port and is color coded red with black stripes. The cable for the torque sensor (2) connects to the TQ port and is color coded yellow with black stripes. Lay the SW and TQ cables next to the manipulator cable, toward the controller, taking care not to twist or tangle the cables.

After all cables are properly connected on the robot side, make the connections to the YRC1000micro controller (refer to the controller panel diagram below). The manipulator cable (1) connects to the rectangular -X1 port (color coded white), like the 1BC connector in the previous step. Make sure the connector is firmly seated and the tighten the lever to lock the connector.

Connect the Torque Sensor cable (2) to the -X73 connector on the controller (color coded yellow). Line up the alignment marks and insert gently until you hear the connector click.

Connect the Signal/Lamp cable (3) to the -X74 connector on the controller (color coded red). Line up the alignment marks and insert gently until you hear the connector click.

This completes, the robot-controller connections. Straighten up the cables and route them in a way to prevent tripping hazards or damage to the cables. Disconnect and reconnect the cables to reroute them as necessary.

Connecting the Pendant

Locate the pendant cable (13 or 18). It may already be attached to the pendant. Connect the pendant cable to the -X81 connector on the controller (color coded silver). Line up the alignment marks on the connectors and gently push until hearing a click.

Connecting the AC IN Cable

After the robot and controller are properly connected, connect your power supply. All power connections should be made by a **qualified electrician**.

NOTICE: One end of the power cable has unfinished leads for flexibility in connecting to your power supply.

Spare connectors are provided in the accessory kit for you to fabricate your own cable if the supplied cable is not long enough.

Once the supply end of the AC IN cable (4) is properly constructed and connected to the main power, connect the AC IN cable to the AC IN connector on the controller (color coded orange). This cable is keyed so that it only fits when the contact pins are properly aligned. Rotate the connector slightly if it does not insert. Then tighten the collar on the connector to secure it.

Congratulations!

This completes the basic HC10DT setup. There are additional components that must be connected before your robot can be operated, however, you must review the online training before powering on or operating the robot to prevent damaging the robot. Please register to take the online course at

https://yaskawaacademy.myabsorb.com/#/online-courses/a87d655a-6f3e-4441-a2d7-63a5ea905f18